

Fuzzy Logic and Control

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Recap

THOUGHT

Systems that think
like humans

Systems that think
rationally

BEHAVIOUR

Systems that act
like humans

Systems that act
rationally

HUMAN

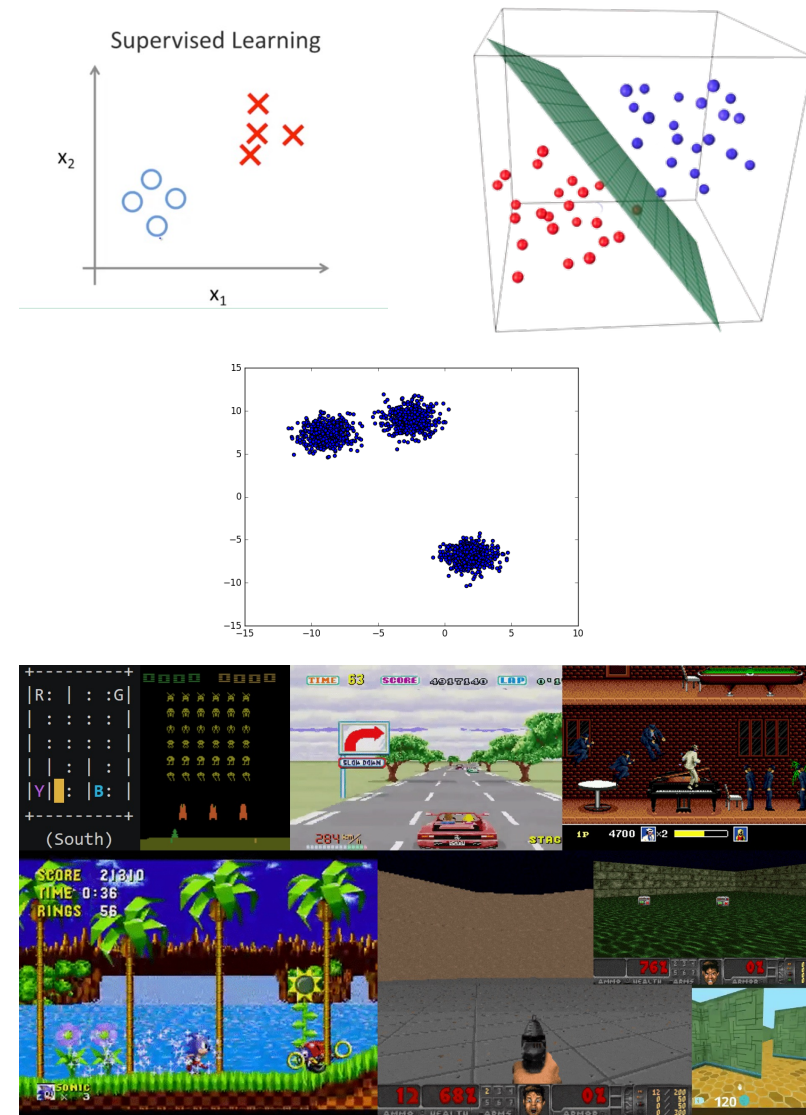
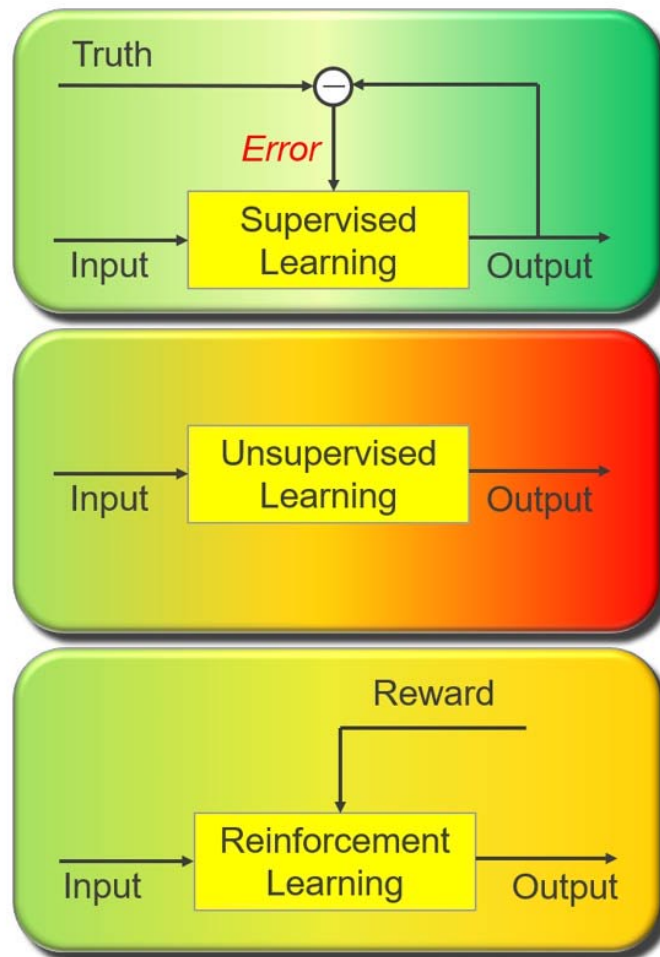
RATIONAL

Recap

Artificial intelligence can be considered under a number of headings:

- Search (includes game playing).
- Knowledge representation.
- Planning.
- Learning.
- Natural language processing.
- Expert systems.
- Robotics.

Recap



Concept: Strong vs. weak AI

- Strong/full AI or artificial general intelligence (AGI):
 - AI has the full range of human cognitive capabilities
 - AI can perform any task that a human can
 - LLMs are getting there
 - (Machines capable of experiencing consciousness)
- Narrow/weak/applied AI:
 - AI able to solve a specific problem
 - Much of the AI used today
 - Example: spam filtering, digital assistants, finding webpages, playing chess, Go etc.

Fuzzy Logic and Control

Fuzzy control provides a formal methodology for representing, manipulating, and implementing a human's heuristic knowledge about how to control a system.

FC book, page 10

An example

You are building a system for a car that should automatically keep a certain distance to any car in front of it on the highway.



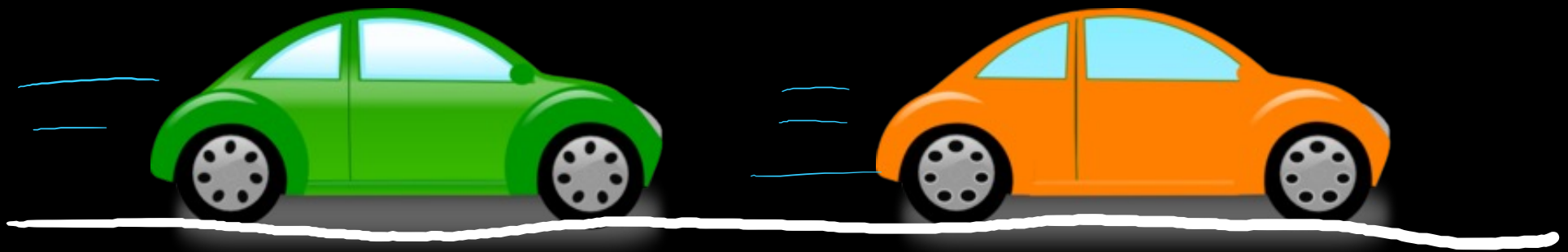
An example

$< 20 \text{ m}$	$\geq 20 \text{ m}$
Brake	Don't brake



An example

< 5 m	5 m – 10 m	10 m – 15 m	15 m – 20 m	≥ 20 m
Brake 100%	Brake 75%	Brake 50%	Brake 25%	Don't brake



Implementation

```
if (fDist < 5.0f)
    fApplyBrake = 100.0f;
else if (fDist < 10.0f)
    fApplyBrake = 75.0f;
else if (fDist < 15.0f)
    fApplyBrake = 50.0f;
else if (fDist < 20.0f)
    fApplyBrake = 25.0f;
else
    fApplyBrake = 0.0f;
```

What about small differences in inputs?

< 5 m	5 m – 10 m	10 m – 15 m	15 m – 20 m	>= 20 m
Brake 100%	Brake 75%	Brake 50%	Brake 25%	Don't apply brake



A solution?

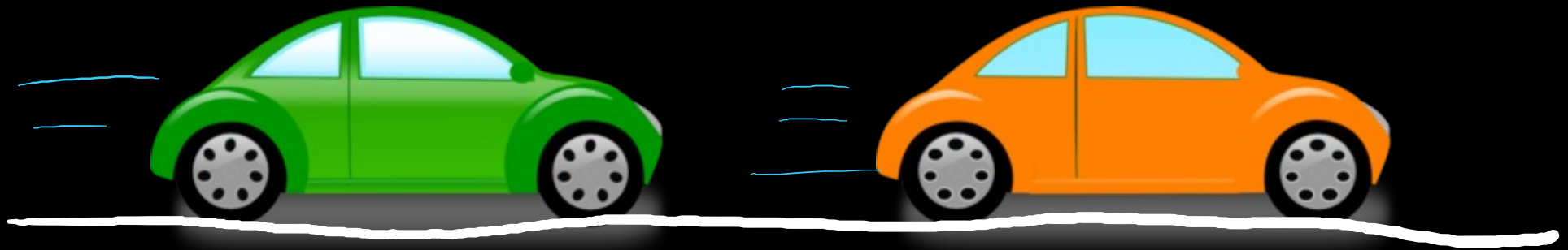


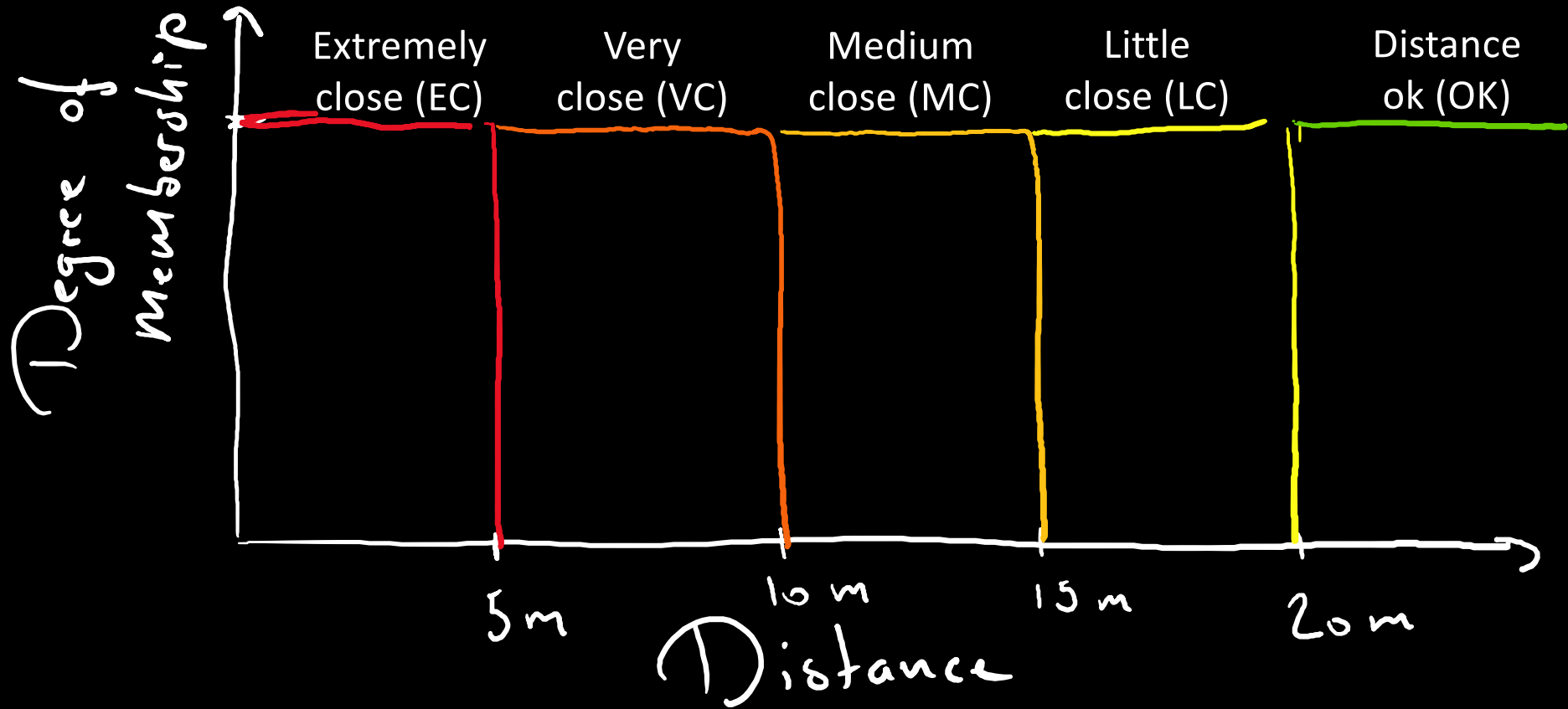
What is the problem?

What is the problem?
Our system uses **crisp**
discrimination between
sets

An example

Extremely Close	Very Close	Medium Close	Little Close	Distance ok
< 5 m	5 m – 10 m	10 m – 15 m	15 m – 20 m	> 20 m
Slam brakes	Brake hard	Brake medium	Brake softly	Don't brake
100%	75%	50%	25%	0%





Boolean logic and indicator/characteristic functions



Definition

The indicator function of a subset A of a set X is a function

$$\mathbf{1}_A: X \rightarrow \{0, 1\}$$

defined as

$$\mathbf{1}_A(x) := \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases}$$

Source: Wikipedia

Many real-world phenomena are not either-or

Examples

- Temperature: { *Very cold, cold, lukewarm, warm, very hot* }
- Property size: { *Small, medium, big* }



small

medium

big

Many real-world phenomena are not either-or

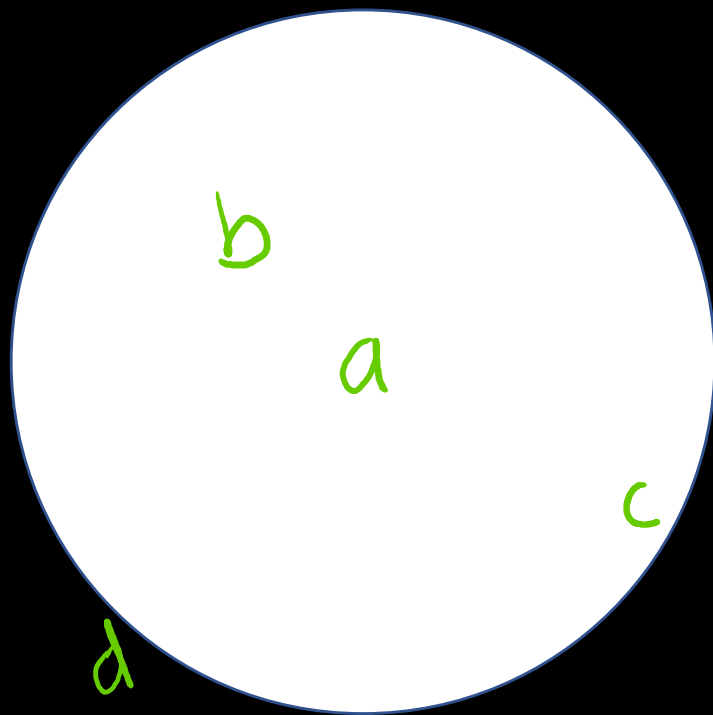
Examples

- Wakefulness: { *Asleep, awake* }

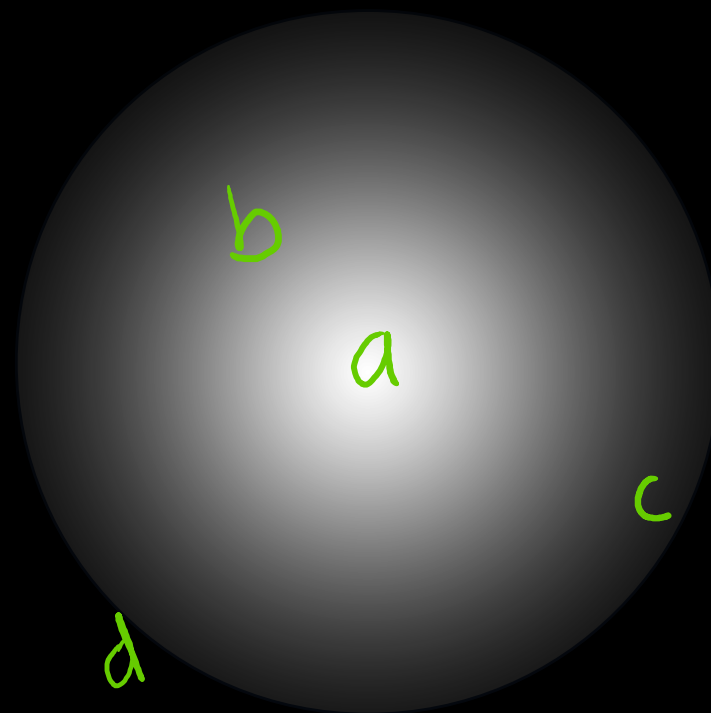
Many real-world phenomena are not either-or

Examples

- Generic error: { Large negative, small negative, zero, small positive, large positive }



Crisp set



Fuzzy set

Indicator function for a crisp set A:

$$I_A(x) \rightarrow \{0,1\}$$

Membership function for a fuzzy set A:

$$\mu_A(x) \rightarrow [0,1]$$

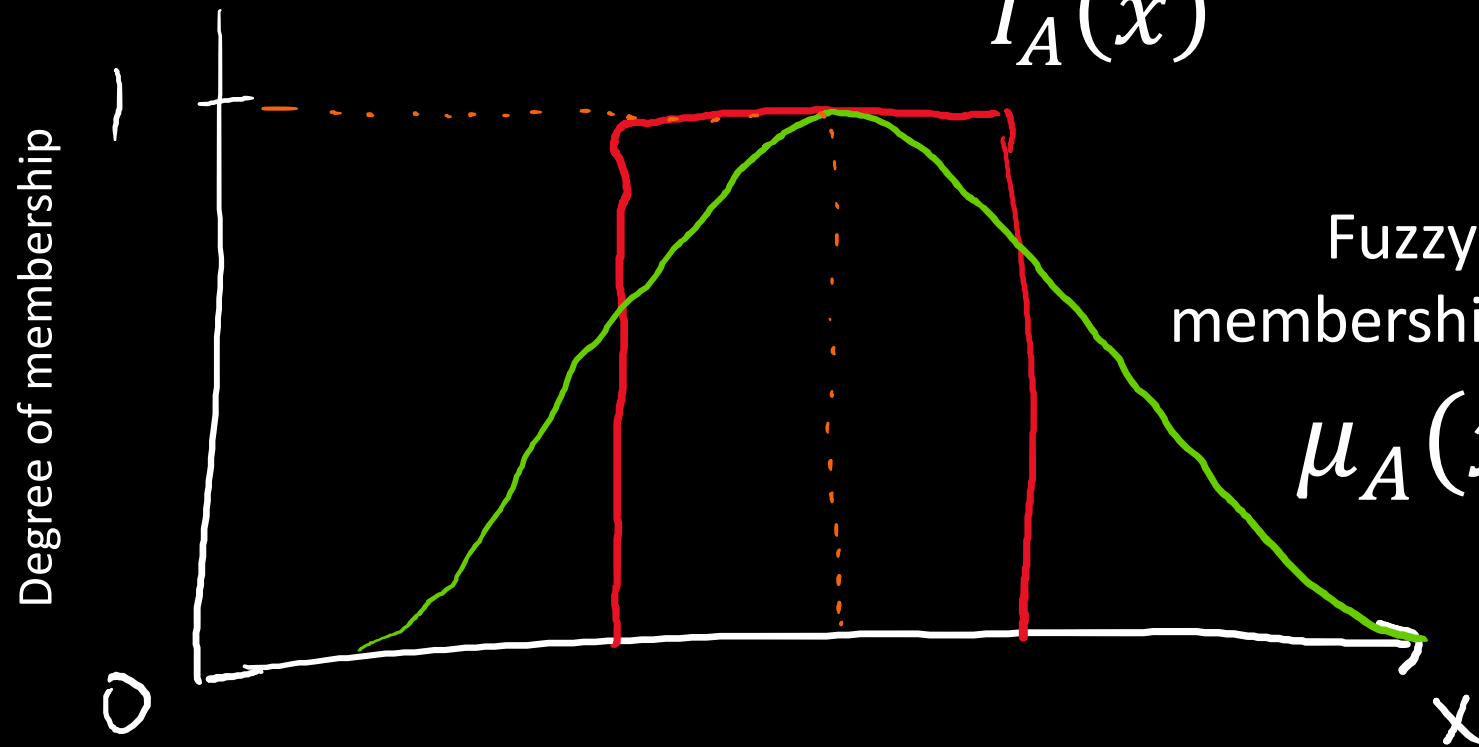
 Membership
degree

Crisp set -
indicator function

$$I_A(x)$$

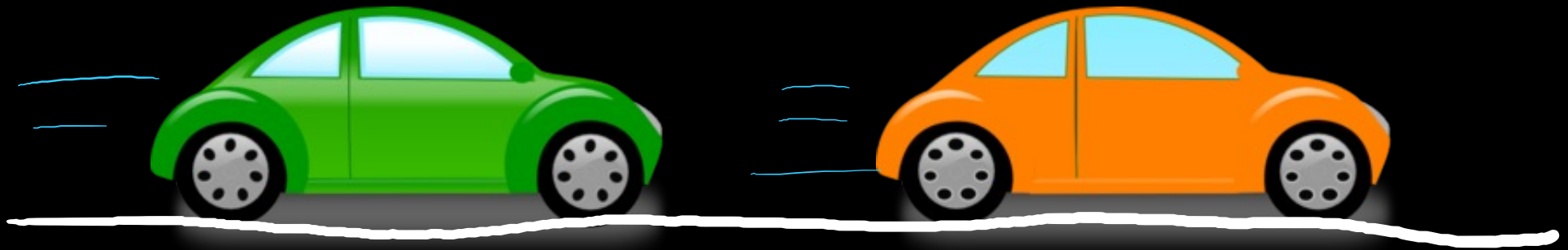
Fuzzy set -
membership function

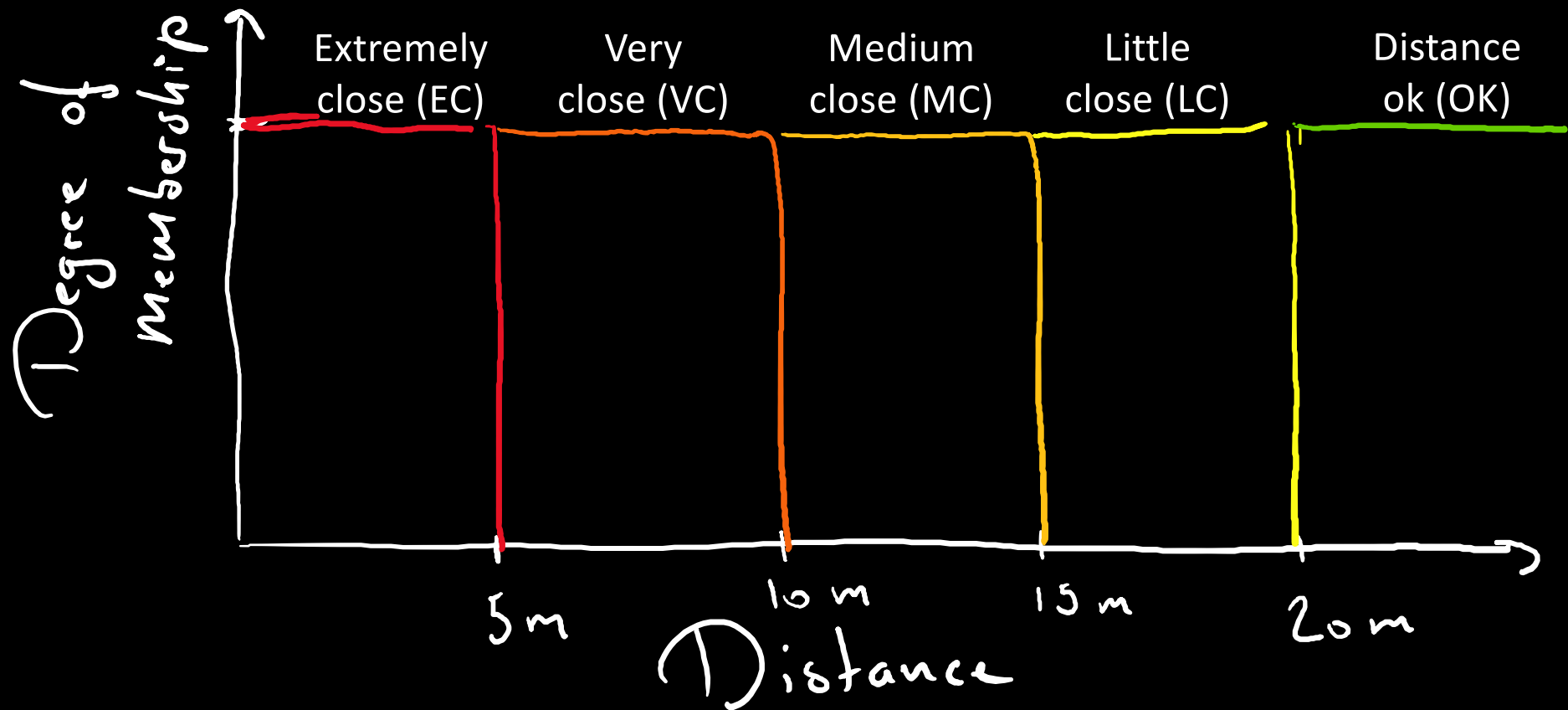
$$\mu_A(x)$$

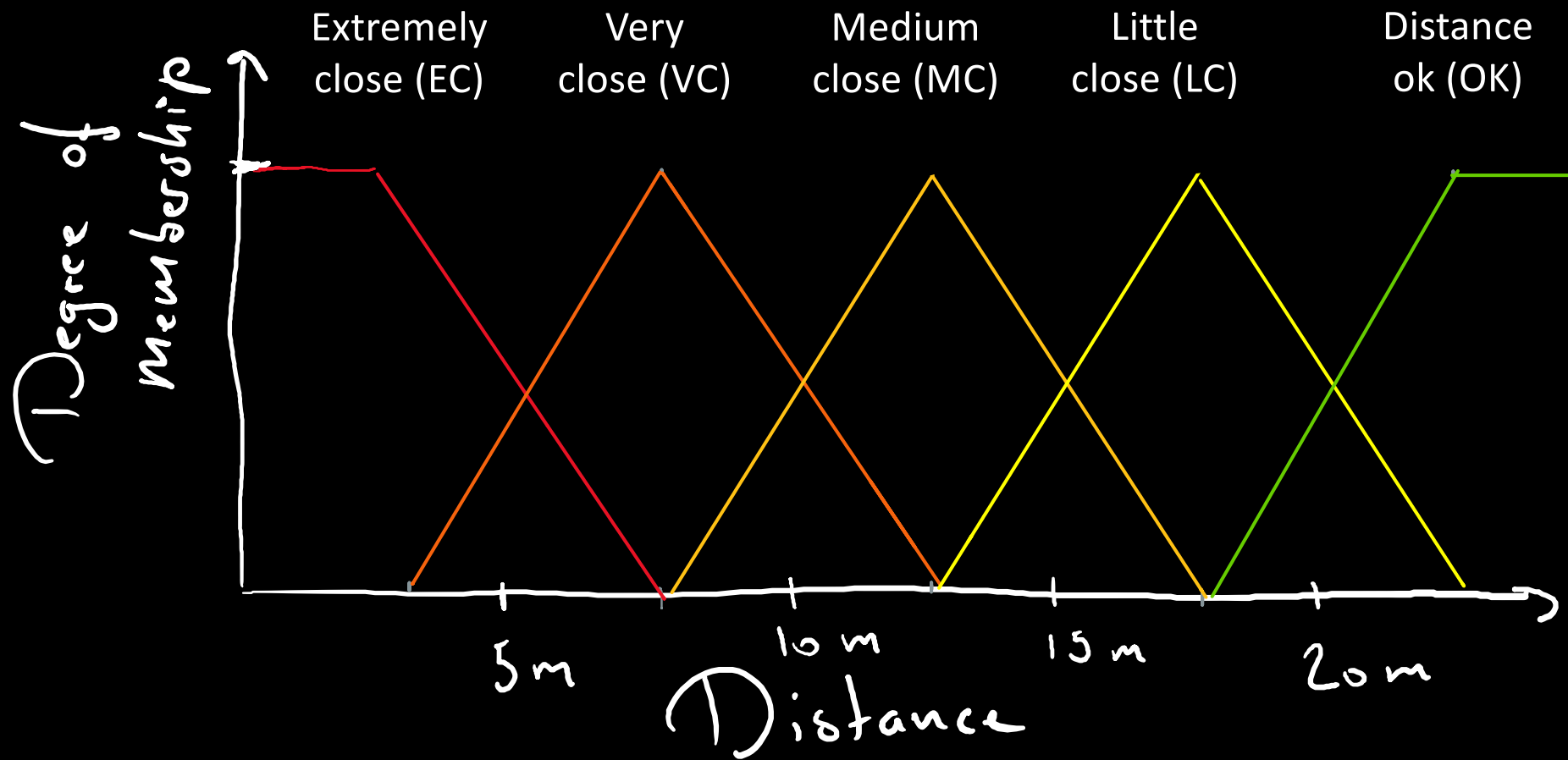


An example

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100%	75%	50%	25%	0%







Our rule-base

if Distance is ExtremelyClose Then Brake is Slam

if Distance is VeryClose Then Brake is Hard

if Distance is MediumClose Then Brake is Medium

if Distance is LittleClose Then Brake is Softly

if Distance is DistanceOk Then Brake is None

Our rule-base

Linguistic rules

if Distance is ExtremelyClose Then Brake is Slam

if Distance is VeryClose Then Brake is Hard

if Distance is MediumClose Then Brake is Medium

if Distance is LittleClose Then Brake is Softly

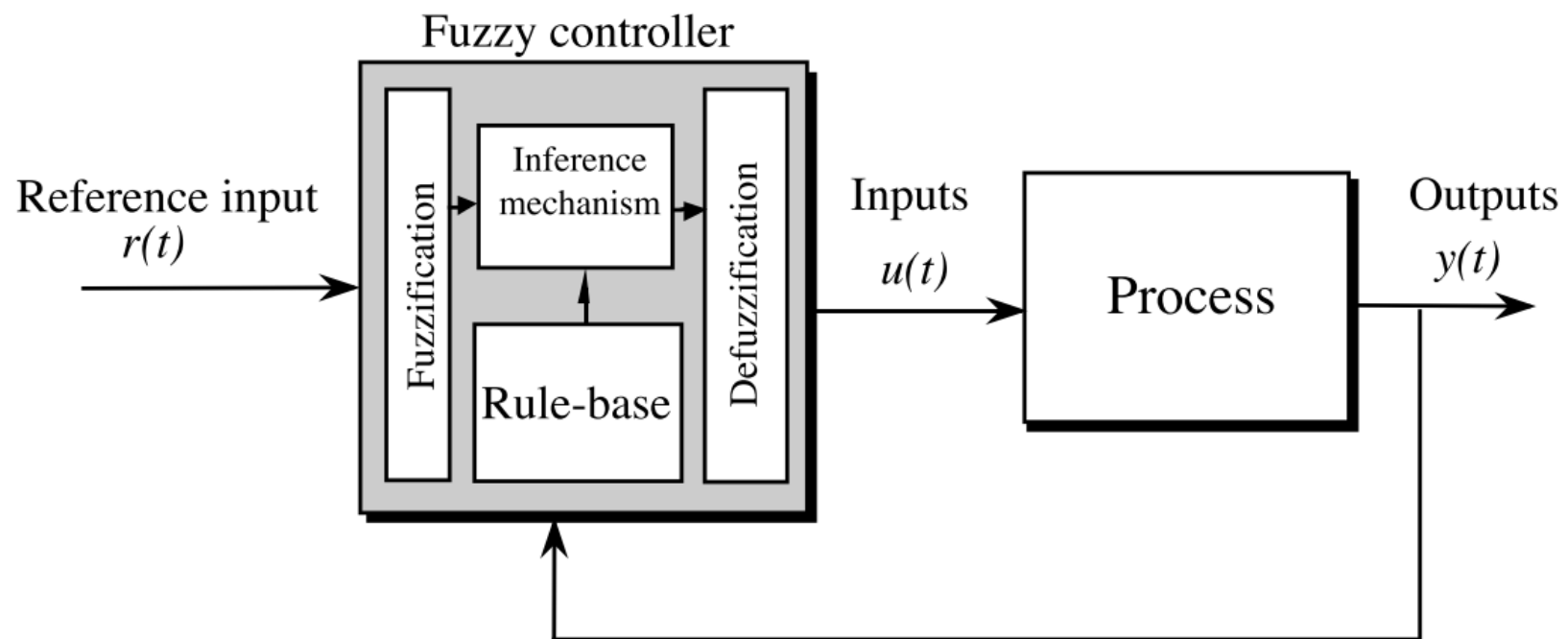
if Distance is DistanceOk Then Brake is None

Linguistic values

→ Linguistic variables

General form of a linguistic rule

If premise/antecedent Then consequent/action



- Fuzzification:
 - Mapping crisp inputs to fuzzy inputs
- Inference
 - Rule matching
- Defuzzification
 - Transforming the fuzzy outputs to crisp outputs

Terminology

- Fuzzy set
- Membership function μ
 - Gives the degree of membership/certainty/truth
- Universe of discourse
 - The range of all possible values for an input to a fuzzy system.
- Rule-base
 - Linguistic rules/variables/values, premises and actions
- Fuzzification, inference, defuzzification

Important: Probability vs. Fuzziness

Probability describes the uncertainty of an event occurrence.

Fuzziness describes event ambiguity.

Whether an event occurs is *random*.

To what degree it occurs/is true is *fuzzy*.

Probability

There is a 50% chance of an apple being in the refrigerator.



Fuzzy

There is a half an apple in the refrigerator.



A fuzzy set A is given by its membership function:

$$\mu_A(x) \rightarrow [0,1]$$

Fuzzy set operations

- Cardinality

- $|A| = \sum_X \mu_A(x)$

- Union

- $\mu_{A \cup B} = \max(\mu_A, \mu_B)$

- Intersection

- $\mu_{A \cap B} = \min(\mu_A, \mu_B)$

- Complement

- $\mu_A^- = 1 - \mu_A$

Example - cardinality

In a class of 10 students (the universal set), three students speak German to some degree, namely Alice to degree 0.7, Bob to degree 1.0, Cathrine to degree 0.4. What is the size of the fuzzy subset A of German-speaking students in the class?

x	0	1	2	3	4	5	6
$\mu_A(x)$	1	0,7	0	1	0,5	0	0,4
$\mu_B(x)$	0,9	0,7	1	0,2	0,8	0,3	0

$$\mu_{A \cup B}$$

$$\mu_{A \cap B}$$

$$\mu_{\bar{A}}$$

$$\mu_{\bar{B}}$$

Note

- We have so far only considered a single input
- Control systems and rules often involve more than one premise
 - How do we compute the degree of certainty for a rule with multiple fuzzy premises?
- Membership functions can take many shapes
- We haven't yet discussed defuzzification in detail

Exercises for today: see [itslearning](#)